DSI-Cohort8-ML-1

Optimizing Bank Marketing Campaigns: Data-Driven Targeting for Term Deposits

Data Sciences Institute — Cohort 8 | Machine Learning Stream

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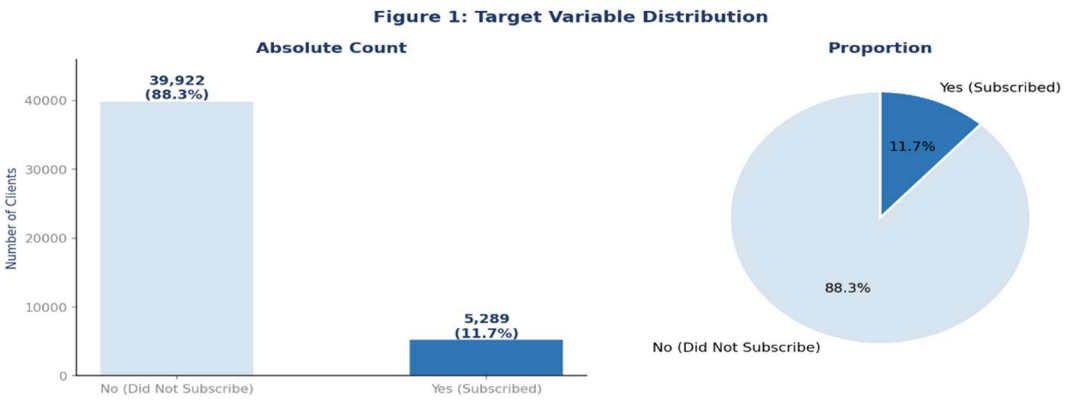
1. Project Overview

The bank's marketing team invested effort in term deposit campaigns which involved substantial operational overhead. However, it yielded a disproportionately low conversion rate, only **11.7%** of subscription rate. This misalignment resulted in significant resource leakage, as the bank continues to engage broad customer segments with low subscription propensity.



2. Goals and Objectives

The goals and objectives of this project is to develop a machine learning model that predicts the likelihood of a client subscribing to a term deposit, leveraging demographic information,

financial profiles, and historical campaign interactions. By identifying high-propensity clients prior to outreach, the bank can enhance campaign efficiency, minimize redundant contacts, and **maximize conversion rates**.

Furthermore, we want to **uncover the key factors** behind a client's decision. These insights will enable the bank to refine its marketing strategies and make data-driven decisions for future campaigns.

3. Methodology & Preprocessing

To ensure robust model performance and data integrity, we implemented the following preprocessing steps.

- Data Leakage Prevention: Removed the duration column to ensure the model only uses information available prior to the call.
- Feature Engineering: Converted the pdays column into a more interpretable binary feature and encoded the target variable into a numeric (0/1) format.
- Data Sanitization: Stripped whitespace from string columns to ensure consistency and removed duplicate records to maintain sample independence.
- Reproducibility: Established a structured data pipeline, preserving the raw data while storing the cleaned data for modeling.

4. Modeling Strategy

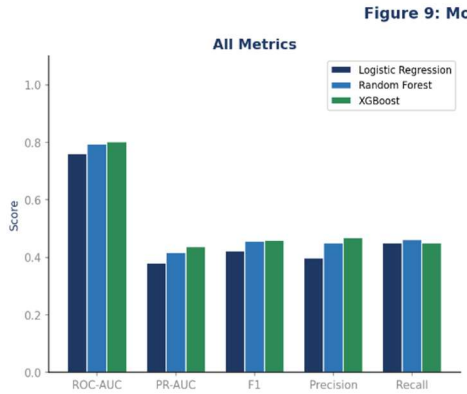
The data was partitioned using **stratified sampling** (**70% Training, 15% Validation, 15% Test**) to maintain class distribution. The project evaluates three distinct models:

- **Logistic Regression (Baseline):** Establishes a performance floor and provides initial transparency.
- **Random Forest (Intermediate):** Captures non-linear relationships and provides feature importance rankings.
- **XGBoost (Primary):** A state-of-the-art framework for tabular data, expected to deliver the highest predictive performance.

5. Model Evaluation

To boost marketing efficiency, the bank will shift from calling all customers to a data-driven approach using the **XGBoost model**. Our evaluation shows that XGBoost is the most precise model, achieving the highest **ROC-AUC (0.8024)** and **Precision (0.4691)**. This high precision is

crucial for the bank, as it minimizes "**false positives**" ensuring that marketing teams spend less time on uninterested leads.

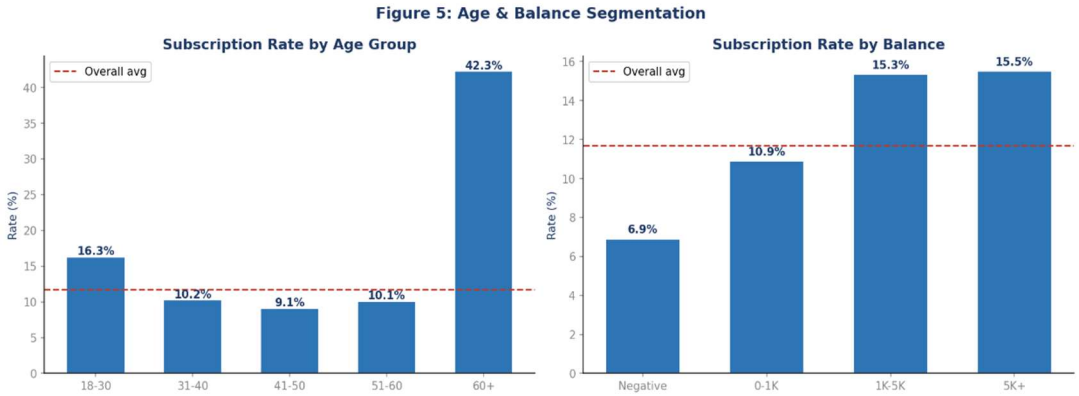


	ROC-AUC	PR-AUC	F1	Precision	Recall
Logistic Regression	0.7599	0.3808	0.4222	0.3976	0.4502
Random Forest	0.7947	0.4166	0.4558	0.4502	**0.4615**
XGBoost	**0.8024**	**0.4368**	**0.4595**	**0.4691**	0.4502

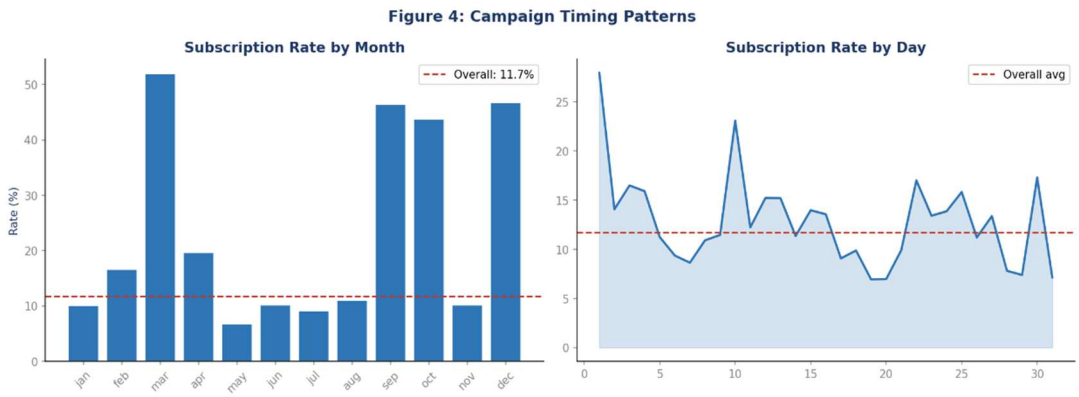
6. Key Findings

The exploratory data analysis reveals specific high-conversion segments:

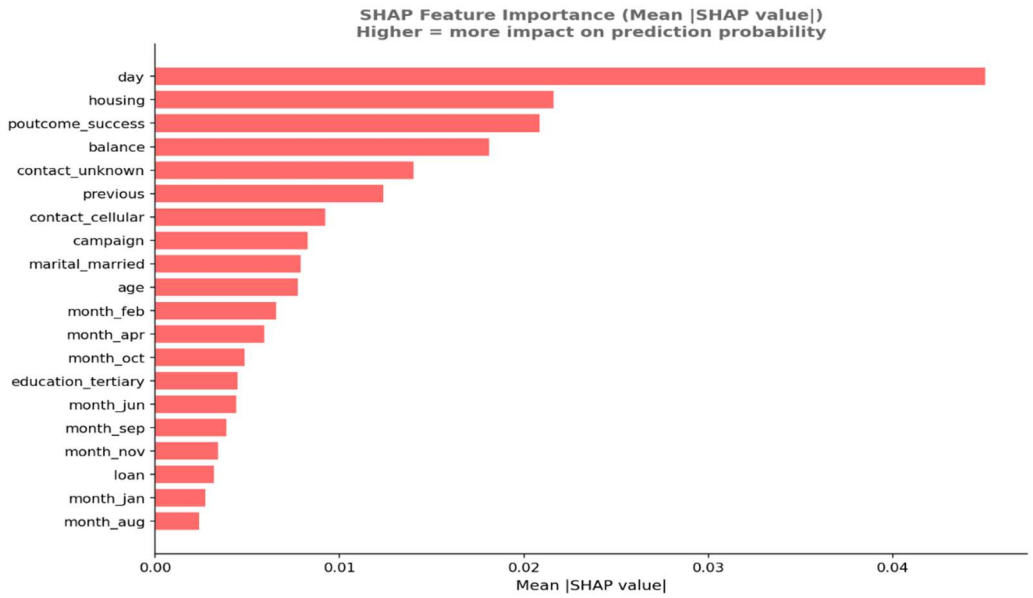
- Demographics Propensity: **Seniors** (age 60+) are the most receptive group, boasting a remarkable **42.3%** subscription rate, reflecting the demand of low-risk preservation products. Additionally, **young adults** (age 18-30) also show strong engagement at **16.3%**, indicating early-stage financial planning.
- Financial Health: Clients with **balances over 5K** subscribe at more than double the rate (15.5%) of those with negative balances (6.9%).



- **Seasonality**: Subscription success rates exhibit distinct seasonal peaks in March, September, October, and December. These periods likely coincide with quarterly and year-end financial planning cycles.



- Previous Success: The strongest indicator of a "Yes" is a **successful previous** marketing outcome, which boasts a remarkable 64.7% conversion rate. This indicates high customer retention and product affinity among clients who have previously engaged with the bank's offerings.
- Key factors behind client's decision: As shown from the SHAP Feature, **day of the month, housing, previous success**, and balance are the key features that impact on model predictions.



7. Business Impact & Actionable Strategy

The transition from a "**blanket**" calling strategy to a model-driven approach generates significant ROI through two primary levers:

Efficiency Gains (Cost Reduction)

- By deploying the XGBoost model, the bank can identify a high-probability subset of clients with superior precision. Rather than exhausting resources on the entire base

of 45,000+ leads, the bank can strategically target only those identified by the model. This predictive filtering is estimated to reduce unproductive call volume and operational overhead by **40-50%**, while still capturing the vast majority of potential subscribers and significantly mitigating "**call fatigue**".

Targeted Revenue Growth

- High-Value Segment Prioritization: Targeting efforts should be focused on **Seniors** (Age 60+) and clients from **tertiary education**. These segments show significantly higher conversion rates.
- Strategic Timing: Reallocate marketing budget and staff to March and October. The data indicates a strong **seasonal effect**, with clients being nearly 5x more likely to subscribe in March compared to May.
- Product Positioning: Given that **high-balance** clients are more likely to subscribe, rather than general savings product, it should be cross-sold as a wealth management tool for existing high-net-worth (**HNW**) customers.

8. Recommendation

Implement the **XGBoost model**, rank all leads by their subscription probability and prioritize outreach to the top two deciles (**Top 20%**). By focusing on high -propensity segments, this strategy is projected to increase the conversion-per-call ratio by at least **3x** compared to a randomize calling baseline.